Mathematical model

When the 20 sick explorers arrived, they infected the 980 Htam people on Ledom Island with the flu. You want to investigate how the resulting flu epidemic might have developed. So, now you construct a mathematical model. (Surprise!)

First define the variables:

t: the time in days after the arrival of the explorers in the 15th century.
$F(t)$: the number of people infected with the flu.
$X(t) = \dfrac{F(t)}{1000}$: the fraction of the infected people relative to the entire population.

Just after the explorers arrived, 2 percent of the population on Ledom Island was sick with the flu, so $X(0) = 0.02$.

In this model, we will assume that the flu epidemic spreads so quickly, that everyone has been infected before the first people get better again.

Exercise A

1/1 point (graded)
What influences the number of infections? First, infection can only occur when a person with the flu infects a person without the flu, obviously. For that, they have to be near to each other for some time. We will call that meeting an `encounter'.

Encounters of two people on the island can be between two healthy people, between two sick people, and between a healthy and a sick person. Someone new can only be infected in the last type of encounters.

When we `sample' the people on the island, the chance that the person is sick is X. The chance that the person is healthy is $(1 - X)$. Now calculate the chance that an encounter between two people is between a healthy and a sick person. Remember, the population of Ledom Island is large.

What is the probability that a encounter between two people is between an infected person and a healthy person?

- X

- $X\,(1-X)$
- $2\,X\,(1-X)$ ✔
- $1-X$

Answer
Correct:
Adding the chances for pairs to be sick/sick, healthy/healthy, sick/healthy (and healthy/sick) gives: $X^2+(1-X)^2+2X(1-X)=\ldots$, as it should.

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You have used 1 of 2 attempts

✔ Correct (1/1 point)

To quantify the number of newly infected people, we need two more numbers:

- E: the number of encounters of pairs of people on the island each day.
- P_i: the chance that one encounter between an infected and a healthy person results in infection of the healthy person.

Now you can construct a balance equation.

Consider time interval $[t, t+\Delta t]$.
Define ΔF as the change in the number of infected people during that time interval.

Exercise B

1/1 point (graded)
Which factors should be included in the expression for ΔF?
Check all factors that are necessary.

- [x] $X(t)$ ✔
- [x] $(1-X(t))$ ✔
- [x] E ✔
- [x] P_i ✔

- [x] Δt ✓
- [ ] $F(0) = 20$ or $X(0) = 0.02$

✓

Answer
Correct:
More people with the flu cause more infections.
Indeed, more people that have not yet had the flu provide a larger breeding ground for the virus, and thus more people will be infected.
Yes, if people meet more often, infection will spread quicker.
Yes, the easier people infect each other during an encounter, the higher the number of infections.
Right, the longer the time interval, the higher the number of infections in it.
Indeed, $F(0)$ (and $X(0)$) are only relevant for the first interval $[0, \Delta t]$.

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Answers are displayed within the problem

Exercise C

1/1 point (graded)
Assume that the total number of inhabitants of the island stays the same for some time and that E and P_i are constant.

The balance equation for ΔF is:

- ( ) $\Delta F = -2EP_i\, X(t)\Delta t$
- ( ) $\Delta F = +2EP_i\, X(t)\,(1 - X(t))$
- ( ) $\Delta F = -2EP_i\, X(t)\,(1 - X(t))\,\Delta t$
- (•) $\Delta F = +2EP_i\, X(t)\,(1 - X(t))\,\Delta t$ ✓

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Answers are displayed within the problem

Exercise D

1/1 point (graded)

Derive a differential equation for $F(t)$ in the usual way. Then carefully substitute $F(t) = 1000X(t)$. The differential equation for $X(t)$ then becomes

$$\frac{dX}{dt} = k\, X(t)\,(1 - X(t))\,.$$

The relation between parameters k and the other constants is:

- $k = \frac{EP_i}{500000}$
- $k = \frac{EP_i}{500}$ ✔
- $k = 2EP_i$
- $k = 2000EP_i$
- $k = 2000000EP_i$

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Exercise E

1/1 point (graded)

What is the unit of k?

- $[k] = (person)^{-1}(day)^{-1}$
- $[k] = (day)^{-1}$ ✔
- $[k] = person$
- $[k] = 1$ (dimensionless)

Answer

Correct: X is a fraction, so dimensionless, but $\frac{dX}{dt}$ is a velocity: the change of the fraction per day.

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Answers are displayed within the problem

On the next page, you will investigate the solutions of the differential equation for $X(t)$.